

Science Through Advanced Research, Learning, Application, and Building

Course Proposal (1-Year Timeline)

1. Course Objectives (1-Year Timeline)

1. **Understand the Ethical Considerations in Research**
 - **Objective:** Students will gain a solid understanding of the ethical principles involved in scientific research, including responsible conduct, research ethics, and the importance of integrity in scientific inquiry.
 - **Outcome:** By the end of Unit 1, students will be able to identify ethical considerations relevant to their projects and complete necessary documentation (e.g., IRB forms) to ensure compliance with ethical research practices.
2. **Develop a Comprehensive Research Question or Engineering Goal**
 - **Objective:** Students will learn how to identify, articulate, and refine a scientific question or engineering goal that is relevant, novel, and feasible within the constraints of the course.
 - **Outcome:** By the end of Unit 2, students will produce a well-defined research question or engineering goal that demonstrates clarity, focus, and alignment with scientific inquiry or engineering design principles.
3. **Demonstrate Proficiency in Experimental Design and Iteration**
 - **Objective:** Students will learn the fundamentals of designing experiments, including identifying variables, establishing controls, and iterating based on preliminary results to improve project outcomes.
 - **Outcome:** By the end of Unit 3, students will have designed, implemented, and refined their experiments through multiple iterations, demonstrating an understanding of the scientific method and the importance of experimentation in research.
4. **Build Advanced Skills in Data Analysis**
 - **Objective:** Students will gain proficiency in various data analysis techniques, including descriptive statistics, inferential statistics, and data visualization methods, enabling them to analyze and interpret experimental data effectively.
 - **Outcome:** By the end of Unit 4, students will be able to apply appropriate statistical methods to their data sets, draw valid conclusions, and create visual representations of their findings that enhance understanding and clarity.
5. **Present Findings in Both Written and Oral Formats**
 - **Objective:** Students will develop strong scientific writing and communication skills, focusing on how to effectively convey their research process and findings to diverse audiences through reports and presentations.
 - **Outcome:** By the end of Unit 5, students will produce a formal research report and deliver an engaging oral presentation that communicates their work clearly, effectively, and professionally.
6. **Engage in Peer Review and Collaborative Processes**
 - **Objective:** Students will learn the importance of collaboration in science by participating in peer review sessions, providing and receiving constructive feedback, and working together on project development.
 - **Outcome:** By the end of Unit 6, students will demonstrate the ability to critically evaluate their peers' work and improve their own projects based on feedback, fostering a collaborative and supportive scientific community.
7. **Compile a Professional Research Portfolio**
 - **Objective:** Students will document their research journey, including project proposals, experimental designs, data analyses, reports, and reflections, in a comprehensive portfolio that showcases their work.
 - **Outcome:** By the end of the course, students will have created a polished research portfolio that highlights their scientific achievements, skills, and growth, serving as a valuable asset for college applications and future opportunities.

2. Course Schedule (1-Year Timeline)

This schedule retains the essential research phases from the two-year model but compresses each phase to focus on critical elements and efficient progression through experimentation and analysis.

1st Semester: Project Selection to Data Collection

1. **Phase 1 (Project Selection and Initial Research) - Weeks 1-2**
 - Students brainstorm ideas, consult with mentors, and conduct preliminary research.
 - Formulate a research question or engineering goal and begin a project journal.
2. **Phase 2 (Planning and Hypothesis Development) - Weeks 3-4**
 - Develop a hypothesis or design criteria, outline an experimental plan, and complete initial safety forms.
3. **Phase 3 (Pilot Experimentation) - Weeks 5-6**
 - Conduct initial testing or prototypes to refine the methods and adjust plans.
4. **Phase 4 (Data Collection) - Weeks 7-16**
 - Carry out the main experiments, ensuring thorough data collection and documentation.

2nd Semester: Data Analysis to Project Presentation

1. **Phase 5 (Data Analysis and Interpretation) - Weeks 17-20**
 - Organize data, perform statistical analyses, and interpret findings.
2. **Phase 6 (Report Writing and Presentation Preparation) - Weeks 21-28**
 - Create visuals, compile the final report, and design a project board for presentation.
3. **Phase 7 (Presentation and Public Display) - Weeks 29-34**
 - Prepare and deliver a formal presentation, with optional participation in local science fairs.

3. Curriculum Outline (1-Year Timeline)

Unit 1: Introduction to Scientific Research (Weeks 1-2)

- **Topics Covered:** Research ethics, identifying research questions, introductory project planning.
- **Activities:** Brainstorming, mentor consultations, project selection, journaling setup.

Unit 2: Hypothesis and Research Plan Development (Weeks 3-4)

- **Topics Covered:** Hypothesis formation, experimental variables and controls, material identification.
- **Activities:** Research planning, resource organization, completing relevant safety and approval forms.

Unit 3: Pilot Testing and Data Collection (Weeks 5-16)

- **Topics Covered:** Preliminary testing, full experimentation, systematic data recording.
- **Activities:** Conduct pilot tests to refine the project approach, then collect and document data.

Unit 4: Data Analysis and Interpretation (Weeks 17-20)

- **Topics Covered:** Statistical methods, data visualization, analysis, and drawing conclusions.
- **Activities:** Analyze collected data, interpret results, and prepare initial findings.

Unit 5: Scientific Reporting and Presentation (Weeks 21-28)

- **Topics Covered:** Scientific writing, visual representation of data, public speaking techniques.
- **Activities:** Write the research report, create a project board, and prepare for oral presentations.

Unit 6: Project Presentation and Reflection (Weeks 29-34)

- **Topics Covered:** Presentation skills, reflection on research process, optional competition preparation.
- **Activities:** Present projects to peers, school community, or at a science fair; conduct self-reflection and final reporting.

This condensed, year-long schedule allows students to experience the full research cycle, focusing on core experimental skills while preparing them to present their work in professional formats.

4. Course Syllabus (1-Year Timeline): Science Through Advanced Research, Learning, Application, and Building (STARLAB)

Instructor:

Grade Level: 11th – 12th grade

Prerequisites: Completion of introductory science courses (Biology, Chemistry, or Physics) or instructor approval
Course Duration: 1 Year

Course Description

This course provides high school students with a unique opportunity to conduct rigorous, long-term scientific research in a chosen field of interest. With the support of a mentor-facilitator, students will design, execute, and report on self-directed research projects. Through these projects, students will acquire valuable skills in experimental design, data analysis, and scientific communication, creating a competitive portfolio for future academic and professional pursuits.

Course Objectives

By the end of this course, students will be able to:

- Develop and refine a scientific question or engineering problem.
 - Design and conduct an experiment or engineering solution with appropriate control and rigor.
 - Collect, analyze, and interpret data to make informed conclusions.
 - Present findings through written reports, visual media, and oral presentations.
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Units and Weekly Breakdown

Unit 1: Introduction to Scientific Research and Experimental Design (Weeks 1-2)

- **Topics Covered:**
 - Fundamentals of scientific inquiry and research ethics
 - Basics of experimental design: variables, controls, and replicability
 - Brainstorming and selecting research topics
- **Activities:**
 - Guided brainstorming sessions and literature review
 - Development of a research question and discussion with peers and mentors
- **Learning Outcomes:**
 - Students will identify a feasible, original research topic with a clearly defined question or hypothesis.

Unit 2: Project Planning and Hypothesis Development (Weeks 3-4)

- **Topics Covered:**
 - Developing testable hypotheses or engineering design criteria
 - Planning experimental procedures and identifying necessary materials
 - Safety and ethics, including form completion (e.g., IRB, safety forms)
- **Activities:**

- Research planning workshop
- Safety protocol review and approval processes (as required)
- **Learning Outcomes:**
 - Students will produce a research plan with a detailed timeline, materials list, and risk assessment if necessary.

Unit 3: Pilot Testing and Data Collection (Weeks 5-16)

- **Topics Covered:**
 - Pilot testing: refining methods, gathering preliminary data
 - Documentation of procedures and findings in a project journal
 - Techniques for systematic data collection and measurement
- **Activities:**
 - Conducting preliminary trials and adjusting experimental setup based on findings
 - Regular entries in a project journal, documenting observations and adjustments
- **Learning Outcomes:**
 - Students will conduct pilot tests, adjust their methodologies as needed, and begin systematic data collection.

Unit 4: Data Analysis and Interpretation (Weeks 17-20)

- **Topics Covered:**
 - Data organization, statistical analysis, and error assessment
 - Data visualization: charts, graphs, and other visuals for analysis
 - Comparison of findings to initial hypothesis or design goals
- **Activities:**
 - Statistical analysis workshop (basic to advanced methods, depending on project needs)
 - Data interpretation and discussion of preliminary findings
- **Learning Outcomes:**
 - Students will analyze and interpret their data, assessing whether results support the hypothesis or meet design criteria.

Unit 5: Scientific Reporting and Presentation Preparation (Weeks 21-28)

- **Topics Covered:**
 - Structure of scientific reports (introduction, methods, results, discussion, conclusion)
 - Creating visuals that accurately represent data
 - Techniques for effective oral presentations

- **Activities:**
 - Drafting and peer-reviewing research reports
 - Designing visual aids for project boards (graphs, tables, images)
 - Practice presentations with feedback sessions
- **Learning Outcomes:**
 - Students will produce a well-organized scientific report and prepare for a public presentation.

Unit 6: Project Presentation and Reflection (Weeks 29-34)

- **Topics Covered:**
 - Presentation delivery, engaging with an audience, and responding to questions
 - Reflection on the research process: challenges, solutions, and personal growth
 - Optional guidance on submitting projects to science fairs or journals
 - **Activities:**
 - Final presentation to peers, teachers, or at a school fair
 - Group reflection session on the course and individual projects
 - For interested students, preparation for submitting research to competitions or publications
 - **Learning Outcomes:**
 - Students will confidently present their findings, demonstrating understanding and articulation of their research.
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Assessment and Grading

Component	Weight
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Project Proposal	15%
Research Journal/Documentation	20%
Mid-Project Progress Report	15%
Data Analysis and Interpretation	20%
Final Research Report	15%
Oral Presentation	15%

Resources and Tools

- **Primary Resources:** Access to lab equipment, relevant software for data analysis (e.g., Google Sheets, Excel, or statistical software), presentation software, and a research journal
 - **Supplementary Texts and Websites:**
 - *The Science Fair Project Guide* by The Society for Science & the Public
 - Online resources: SciFy Initiative, Rules Wizard (ISEF), local university databases
 - **Mentorship and Collaboration:** Regular access to mentor meetings, peer-review sessions, and guest speakers from scientific and engineering fields as available
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Expectations

Students are expected to:

- Engage actively in each phase of research, maintaining consistent documentation.

- Collaborate respectfully with peers and mentors, providing and receiving constructive feedback.
- Adhere to project deadlines and demonstrate a commitment to revising and refining their work based on data and peer insights.

Accommodations

Individualized support will be available to ensure accessibility for students with diverse needs. Adjustments in pacing, alternative materials, or additional mentor guidance will be provided as needed.

5. NCCRS-S Standards Alignment

This proposed science course would align with and enhance the Nebraska College and Career Ready Standards for Science (NCCRS-S) by addressing core components like inquiry-based learning, scientific methodology, and cross-disciplinary connections. Here's how this integration works:

1. Alignment with Science and Engineering Practices

- The course emphasizes practices central to the Nebraska standards, such as asking questions, defining problems, developing models, planning investigations, analyzing data, and interpreting results. These skills are foundational in both the NCCRS-S and the course's curriculum, allowing students to engage in practices similar to those used by professional scientists. Activities like iterative experimentation and hypothesis testing provide students hands-on, in-depth experiences that go beyond surface-level understanding.

2. Support for Disciplinary Core Ideas Across Sciences

- By encouraging students to choose projects from a wide range of scientific domains, this course naturally incorporates core ideas from physical science, life science, earth and space science, and engineering as outlined in the standards. For example, a project on high-altitude weather balloons might incorporate physical science (forces and motion) and earth science (atmospheric layers and weather patterns), aligning well with standards in these domains while reinforcing interdisciplinary learning.

3. Crosscutting Concepts Emphasis

- Nebraska's standards include crosscutting concepts like patterns, cause and effect, scale, energy, systems, and stability. Through projects, students will explore these concepts in real-world applications. For instance, a project on ecosystem impacts of urban development could cover cause and effect and systems thinking. Crosscutting concepts allow students to connect learning across fields, enhancing their ability to apply science to complex, interconnected challenges.

4. Focus on College and Career Readiness

- The course develops college and career readiness skills, particularly through sustained, independent research. Students develop competencies in critical thinking, quantitative analysis, and technical writing, which are all critical for success in both post-secondary education and 21st-century careers. These align directly with Nebraska's focus on preparing students for diverse post-secondary pathways, from academia to industry.

5. Engineering and Design Integration

- The Nebraska standards emphasize engineering as a core part of science education. This course mirrors that focus, especially in phases involving prototyping and iteration, allowing students to experience the engineering design process. By guiding students through refining and testing their designs, the course aligns well with Nebraska's engineering standards, which emphasize the iterative process of defining, designing, and optimizing solutions.

6. College Readiness and 21st-Century Skill Development

1. **Research Methodologies**

- **Skill Development:** Students learn to formulate research questions, design experiments, and conduct thorough literature reviews. They will gain proficiency in using academic databases and resources.
- **College Readiness:** This prepares students for college-level research and coursework, where critical inquiry and analytical thinking are essential. They will be equipped to undertake independent studies, whether in scientific disciplines or humanities.

2. **Data Analysis and Interpretation**

- **Skill Development:** Students will engage with statistical software and learn various data analysis techniques, including descriptive and inferential statistics, data visualization, and trend analysis.
- **College Readiness:** Proficiency in data analysis is increasingly crucial in higher education, especially in STEM fields. Students will be able to tackle complex datasets, which is often a requirement in college courses.

3. **Scientific Writing and Reporting**

- **Skill Development:** Through structured assignments and peer reviews, students will develop their scientific writing skills, focusing on clarity, structure, and adherence to scientific conventions. They will learn to write research papers and reports that include abstracts, methodologies, results, and discussions.
- **College Readiness:** Strong writing skills are essential in college, where students are expected to produce research papers across various disciplines. This experience will help students articulate their ideas and findings effectively.

4. **Presentation and Communication Skills**

- **Skill Development:** Students will practice oral presentation techniques, including how to engage an audience, use visual aids effectively, and articulate complex ideas clearly.
- **College Readiness:** Public speaking and presentation skills are critical in college, as students often present their work in seminars and group projects. These skills will enhance their ability to communicate their research effectively in academic and professional settings.

5. **Critical Thinking and Problem-Solving**

- **Skill Development:** Through iterative design processes and troubleshooting during experiments, students will enhance their critical thinking and problem-solving abilities. They will learn to analyze situations, make decisions based on data, and adapt their approaches as necessary.
- **21st Century Job Skills:** Employers value employees who can think critically and solve problems creatively. These skills are fundamental across various careers, including engineering, healthcare, business, and technology.

6. **Collaboration and Teamwork**

- **Skill Development:** Students will work with peers and mentors, developing teamwork skills through collaborative projects and peer review processes. They will learn to give and receive constructive feedback, work towards common goals, and share responsibilities.
- **21st Century Job Skills:** Collaboration is a key competency in many professional environments, as most workplaces require teamwork across diverse teams. Strong interpersonal skills will help students navigate group dynamics effectively.

7. **Project Management and Time Management**

- **Skill Development:** Students will plan and execute long-term projects, learning to manage their time effectively, set deadlines, and stay organized. They will use project management tools to track progress and milestones.
 - **21st Century Job Skills:** These skills are essential in any career, as managing projects efficiently can lead to greater productivity and success. Employers often seek individuals who can prioritize tasks and meet deadlines consistently.
8. **Adaptability and Resilience**
- **Skill Development:** Through the iterative process of experimentation and refinement, students will learn to adapt their methods based on findings and setbacks. They will gain resilience by learning to embrace failure as a part of the scientific process.
 - **21st Century Job Skills:** The ability to adapt to changing circumstances and bounce back from challenges is highly valued in the modern workforce, especially in dynamic fields like technology and business.

7. Example Project Ideas

Here are some advanced and novel project ideas designed to offer students deep engagement with experimental design, data analysis, and interdisciplinary research. Each example builds on scientific and engineering principles while encouraging creativity and rigorous methodology:

1. Cubesat Data Analysis and Satellite Communication

- **Project Description:** Students design a simulated or prototype cubesat payload to analyze environmental data from the upper atmosphere. They could work on developing a model cubesat with sensors to monitor temperature, humidity, and radiation levels and design a ground-based receiver for data transmission.
- **Skills Developed:** Systems engineering, data transmission, signal processing, coding (for telemetry software), and environmental monitoring.
- **Potential Outcomes:** Students could gain insight into atmospheric changes, particularly valuable for studying climate patterns. They might also develop communication protocols to simulate real-time data streaming from low Earth orbit satellites.

2. High-Altitude Weather Balloon Experiments

- **Project Description:** Using weather balloons, students can investigate atmospheric conditions, cosmic radiation exposure, or even plant growth under extreme conditions. Sensors could be added to measure pressure, temperature, humidity, and UV radiation at different altitudes.
- **Skills Developed:** Experimental planning for extreme environments, sensor calibration, and remote data collection.
- **Potential Outcomes:** This project would provide students with practical experience in designing for conditions similar to space. They could examine trends in atmospheric conditions and data collection, developing insight into high-altitude research similar to scientific fields like meteorology and astrophysics.

3. Autonomous Environmental Monitoring Robot with Image Recognition

- **Project Description:** Students can create an autonomous robot equipped with sensors and an image recognition system to monitor environmental health in local ecosystems. For instance, it could identify invasive species, measure water quality, or track animal populations using machine learning models for image recognition.
- **Skills Developed:** Robotics, programming (Python, C++), machine learning for image recognition, environmental science.
- **Potential Outcomes:** This project allows students to explore environmental science while applying advanced robotics and machine learning. Insights could help local conservation efforts and teach students about real-world applications of AI in environmental monitoring.

4. Design and Analysis of Efficient Solar-Powered Vehicle Models

- **Project Description:** Teams can design and prototype small-scale solar-powered vehicles that aim to maximize efficiency under various conditions. They could experiment with different types of solar cells, materials for aerodynamics, and gear ratios for optimizing energy use.
- **Skills Developed:** Electrical engineering, energy efficiency analysis, materials science, and data analysis.
- **Potential Outcomes:** This project provides practical experience in renewable energy and could lead to innovations in energy efficiency. Students might even simulate real-world solar vehicle challenges by adjusting environmental factors and optimizing their models.

5. Human-Computer Interaction Study with Biometric Feedback

- **Project Description:** Students could design an experiment to analyze how biometric data (like heart rate, eye movement, or facial expressions) change with different types of stimuli on computers or devices. They can use sensors to measure reactions to colors, images, or sounds to understand human interaction with technology.
- **Skills Developed:** Data collection with biometrics, coding for data visualization, and experimental design in psychology and neuroscience.
- **Potential Outcomes:** Students gain insight into psychology and human-computer interaction (HCI). Their findings could be applied to designing more intuitive user interfaces and creating an understanding of the relationship between technology and stress responses.

6. Advanced Study of Algal Biofuels in a Bioreactor

- **Project Description:** This project involves cultivating algae in a controlled bioreactor to investigate optimal conditions for biofuel production. Students could manipulate variables like light exposure, temperature, and nutrient levels to maximize biomass yield and lipid production for energy conversion.
- **Skills Developed:** Experimental control in biological systems, chemical engineering, biofuel production, data analysis.
- **Potential Outcomes:** Students explore a critical area in sustainable energy and biochemistry, developing skills in managing living systems. They could produce and analyze data relevant to alternative fuel sources, a topic central to modern environmental science.

7. Quantifying Urban Heat Island Effects Using Drones and Thermal Imaging

- **Project Description:** Students can use drones equipped with thermal cameras to map temperature variations across urban and rural areas, studying the urban heat island effect. They could investigate factors that contribute to heat retention, like building materials and vegetation density.
- **Skills Developed:** Remote sensing, geographic information system (GIS) mapping, environmental science, and data interpretation.
- **Potential Outcomes:** By studying urban heat patterns, students could contribute valuable data on urban planning and sustainability, gaining skills in using technology to address environmental issues.

8. Analysis of Microplastic Pollution in Local Waterways

- **Project Description:** Students can collect water samples from local streams, rivers, or lakes and analyze them for microplastic content using filtration methods and microscopy. They could explore the source of the microplastics, types of plastics, and potential impact on local ecosystems.
- **Skills Developed:** Analytical chemistry, environmental science, microscopy, and pollution analysis.
- **Potential Outcomes:** This project highlights local environmental concerns and engages students in a growing area of ecological research. Findings could be shared with community groups or local governments to raise awareness and advocate for environmental action.

9. Study of Antibiotic Resistance in Soil Microbiomes

- **Project Description:** Students collect soil samples from various environments and analyze the bacteria for antibiotic resistance genes using PCR and gel electrophoresis. They can compare resistance levels across locations, such as agricultural versus urban areas, and examine factors contributing to resistance.
- **Skills Developed:** Microbiology, molecular biology, PCR techniques, and data interpretation.
- **Potential Outcomes:** The project explores antibiotic resistance at a community level, providing data valuable for understanding and potentially mitigating resistance spread, aligning with real-world medical research challenges.

10. The Effect of Environmental Stressors on Plant Growth and Gene Expression

- **Project Description:** This experiment involves exposing plants (e.g., *Arabidopsis thaliana* or spinach) to various stressors like drought, salinity, or temperature changes, then measuring growth and gene expression changes through RT-qPCR.
- **Skills Developed:** Plant biology, stress physiology, quantitative PCR, gene expression analysis.
- **Potential Outcomes:** Students learn about plant adaptation mechanisms, potentially informing crop science and climate-resilient agriculture research.

11. Investigating Probiotic Impact on Gut Health Using Model Organisms

- **Project Description:** Using *Caenorhabditis elegans* (roundworm) or *Drosophila melanogaster* (fruit fly) as model organisms, students can examine how probiotics affect gut health by studying gut microbiome changes or monitoring survival and reproduction under stressful conditions.
- **Skills Developed:** Experimental design with model organisms, microbiology, data collection, and analysis.
- **Potential Outcomes:** This project provides insights into microbiome health, demonstrating how probiotics affect host organisms and potentially contributing to dietary science and microbiome research.

12. Synthesis and Characterization of Metal-Organic Frameworks (MOFs) for Gas Adsorption

- **Project Description:** Students can create metal-organic frameworks and test their ability to adsorb gases like CO₂. They can characterize MOF properties using spectroscopic analysis and assess gas adsorption capacity.
- **Skills Developed:** Organic synthesis, analytical chemistry (e.g., FTIR, UV-Vis spectroscopy), and environmental chemistry.
- **Potential Outcomes:** MOFs are significant in environmental chemistry, with implications for carbon capture technologies. Students would gain skills applicable to research in sustainable technology and materials science.

13. Electrochemical Water Splitting for Hydrogen Fuel Production

- **Project Description:** In this project, students use electrolysis to split water into hydrogen and oxygen gases, testing the efficiency of various electrolytes or electrode materials to improve hydrogen yield.
- **Skills Developed:** Electrochemistry, materials science, renewable energy research, and sustainability.
- **Potential Outcomes:** This project explores renewable energy generation and familiarizes students with electrochemical processes, an area of interest in alternative energy fields.

14. Analysis of Environmental Contaminants Using Advanced Spectroscopy

- **Project Description:** Students collect and analyze water or soil samples for pollutants, such as heavy metals or pesticides, using UV-Vis or atomic absorption spectroscopy. They could investigate pollutant levels in urban versus rural environments or before and after rain events.
- **Skills Developed:** Environmental chemistry, sample collection, spectroscopy, and data analysis.
- **Potential Outcomes:** This project provides hands-on experience with analytical techniques commonly used in environmental and chemical analysis, with results potentially relevant to local conservation efforts.

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